Robust annotation transfer across modern genome assemblies and species using synteny-aware coordinate projection

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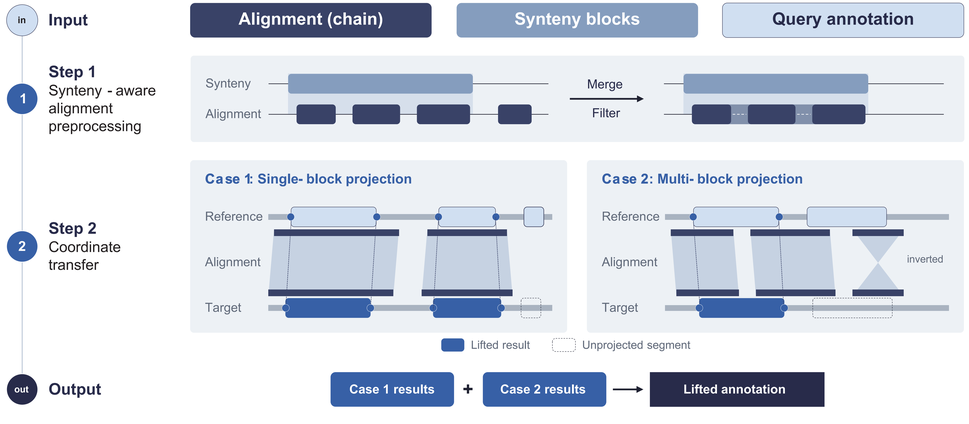
Background & Key Findings

Background Genomic coordinate transfer enables the reuse of existing annotations across updated genome assemblies and facilitates comparative and evolutionary analyses across species.

Approach synLiftOver combines synteny-aware alignment preprocessing with single- and multi-block coordinate projection.

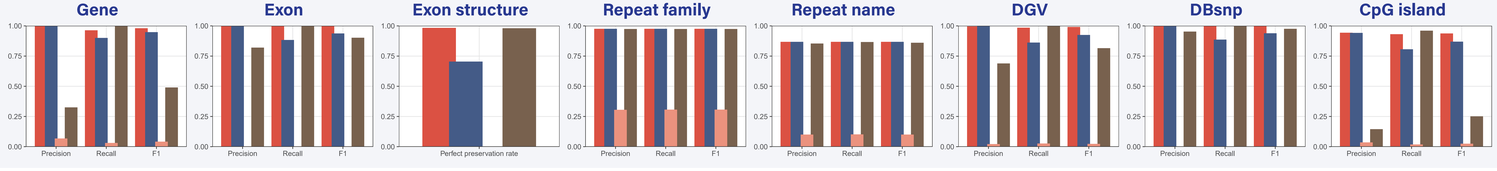
Key Findings
- **Accurate across assemblies**: Reliable annotation transfer across human genome assemblies.
- **Robust across species**: Effective annotation transfer across evolutionarily divergent genomes.
- **Synteny-aware preprocessing**: Filtering and merging offer complementary, context-dependent effects.
- **Multi-block recovery**: Case 2 improves annotation recovery across fragmented alignments.

synLiftOver Workflow

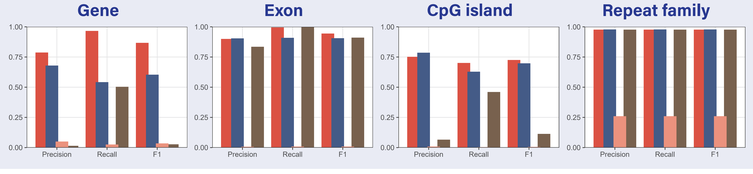


Performance of synLiftOver (Precision, Recall, F1 score)

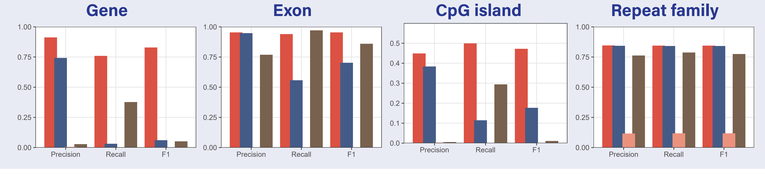
Intraspecies Annotation Transfer (hg19 → hg38)

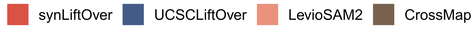


Interspecies Annotation Transfer (hg38 → rheMac10)



Interspecies Annotation Transfer (hg38 → mm39)





Synteny-aware preprocessing reveals context-dependent trade-offs

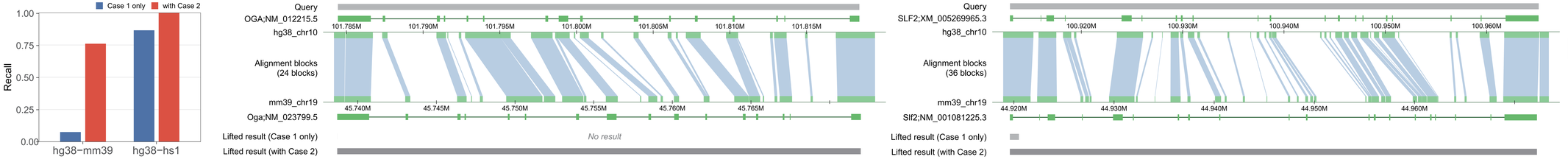
Intraspecies Gene Transfer (hg38 → hs1 (T2T))

Preprocessing	TP / ΔTP	FP / ΔFP	F1
Raw	84,642	405	0.963
Filter	-669	**-102**	0.960
Merge (default)	**+1,634**	+2	**0.973**
Filter + Merge	**+989**	**-97**	**0.969**

Interspecies Gene Transfer (hg38 → mm39)

Preprocessing	TP / ΔTP	FP / ΔFP	F1
Raw	12,826	985	0.838
Filter	-182	**-40**	0.832
Merge (default)	+3	0	0.838
Filter + Merge	-179	**-40**	0.833

Case2:Multi-block projection improves mapping recovery



Acknowledgements

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References

UCSC liftOver: Hinrichs et al., Nucleic Acids Res 2006; CrossMap: Zhao et al., Bioinformatics 2014; LevioSAM2: Chen et al., Nat Methods 2024

Github/ibclab-hub/synLiftOver